

Evolutionary Computation and its Applications

✕

Today's goals

1

▼

Module 1

Fundamental Concepts in Optimization

^

- Welcome
- ✓

Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations

Video • 6 min
- ✓

Specialization and Course Structure

Reading • 10 min
- ✓

Specialization Sample Certificate

Reading • 10 min
- ✓

Stay connected with UofG Online

Reading • 5 min
- ✓

Why Optimization Matters in Engineering and Science

Reading • 5 min

- Module 4 Introduction
- ✓

Introduction to Module 4's Study

Video • 1 min
- ✓

Optimization and Evolutionary Computation in AI

Reading • 5 min

- Module 4-1 Introduction to Optimization
- ✓

Introduction to Optimization: Concepts and Applications

Video • 10 min
- ✓

Introduction to Optimization: Further Applications

Video • 6 min
- ✓

Materials about Optimization

Reading • 10 min

- Module 4-2 Optimization Objectives and Constraints
- ✓

Optimization Objectives and Constraints: Objectives and Constraints

Video • 6 min
- ✓

Optimization Objectives and Constraints: Multiple Objectives

Video • 9 min
- ✓

Materials about Optimization Objectives and Constraints

Reading • 10 min

- Module 4-3 Global and Local Optimization
- ✓

Global and Local Optimization: Concepts

Video • 9 min
- ✓

Materials about Global and Local Optimization

Reading • 10 min

- Module 4-4 Formulate Engineering Problems to Optimization Problems
- ✓

Formulate Engineering Problems to Optimization Problems: Procedures and A Case Study

Video • 11 min
- ✓

Materials about Optimization Problem Formulation

Reading • 10 min

- Module 4-5 Case Studies
- ✓

Case Studies (1)

Video • 3 min
- ✓

Case Studies (2)

Video • 7 min
- ✓

Case Studies (3)

Video • 3 min
- ✓

Case Studies (4)

Video • 46 sec
- ✓

Materials about Case Studies

Reading • 10 min

- Module 4 Assignment
- ✓

Module 4 Quiz

Practice Assignment • Grade: 100%
- ✓

Module 4 Recap

Reading • 5 min
- Integrated Optimization Problem Analysis Dialogue

• 5 min

Module 2

Genetic Algorithm

▼

Module 3

Particle Swarm Optimization

▼

Module 4 Recap

0:05 / 1:15

2x

In this module, you learned how real engineering and scientific problems can be formulated as optimization problems by identifying decision variables, objectives, constraints, search spaces, and different problem types such as single-objective, multi-objective, combinatorial, and continuous optimization.

Reflect on the following questions:

- What were the most important ideas you learned about formulating real-world problems as optimization problems?
- Why do concepts such as feasibility, trade-offs, multimodality, and global versus local optimality matter when solving practical engineering problems?
- Based on what you have learned so far, why might conventional or local optimization methods struggle with complex, multimodal, or constrained problems?
- Genetic algorithms are inspired by natural evolution. Before studying GA in detail, what characteristics do you think an optimization algorithm would need in order to explore a large search space and avoid getting trapped in local optima?

This discussion will help you consolidate the foundational optimization concepts from this module and prepare you to understand **why genetic algorithms work**, not just how to run them, in the next section of the course.

Go to next item

✓ Completed

Like

Dislike

Report an issue